Polymer-Engineered Silicon Nitride Slot Waveguides for Enhanced On-Chip Stimulated Brillouin Scattering

Technische Universität Braunschweig

Jyoti Bej,[1*] Deepanshu Yadav,[1] Thomas Schneider[1]

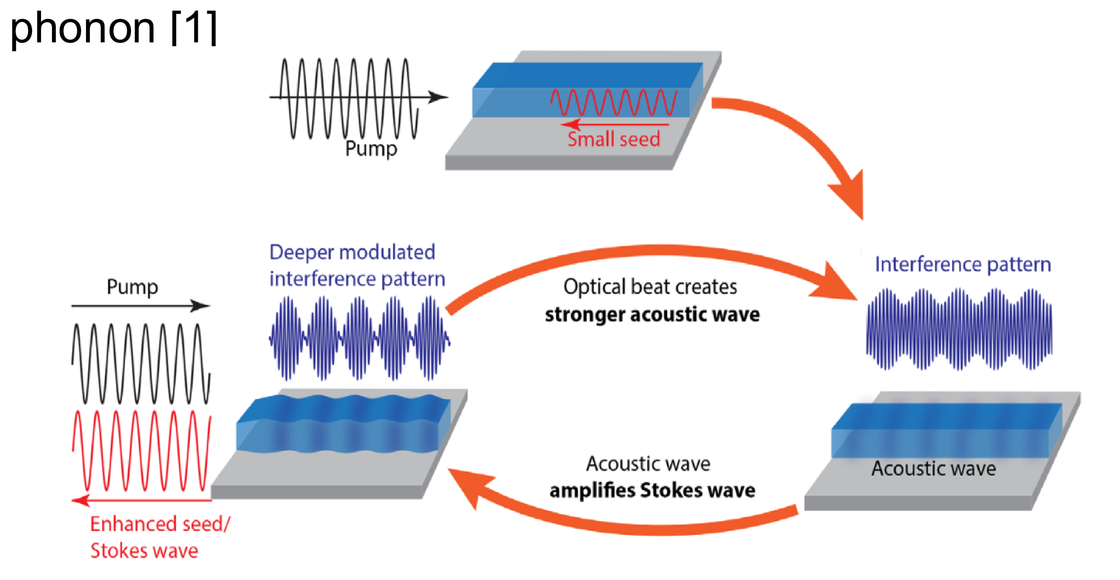
[1] THz-Photonics Group, Institut für Hochfrequenztechnik (IHF), TU Braunschweig

Schleinitzstraße 22, 38106, Braunschweig, Germany

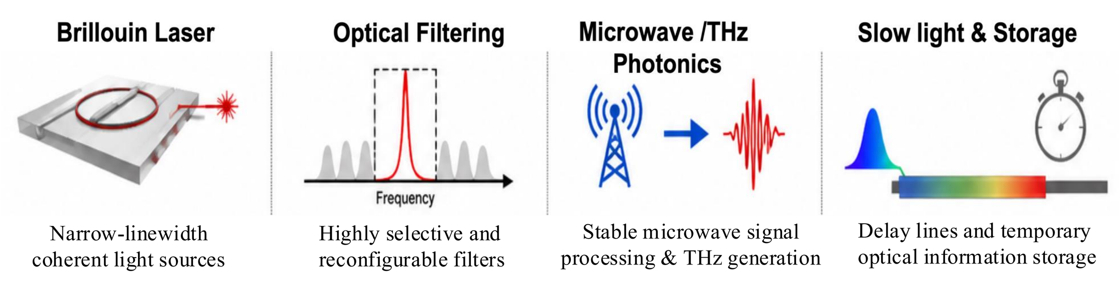
THz PHOTONICS GROUP Braunschweig

1. Motivation

Stimulated Brillouin Scattering (SBS) is a third order nonlinear opto-acoustic interaction between photon and phonon [1]



SBS Applications



2. Si_3N_4 for SBS

Advantages

1. High refractive index
2. Broad transparency
3. Low-loss material
4. CMOS compatibility

Challenges

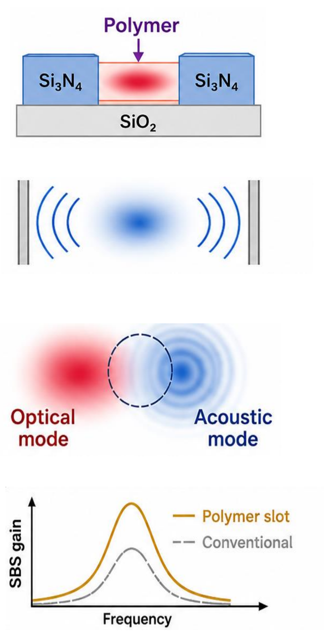
i. Weak acoustic mode confinement

ii. Low photo-elastic response

Limited SBS gain

Goal: Enhance on-chip SBS gain in Si_3N_4 using polymer-engineered slot waveguides through improved acoustic confinement.

3. Why Polymer Slots?



Polymer slot engineering (tailored optical & acoustic properties)

Acoustic confinement tailoring (reduced acoustic leakage)

Stronger opto-acoustic overlap (enhanced photon-phonon coupling)

Higher Brillouin gain (improved Brillouin response)

4. Investigated Slot Materials

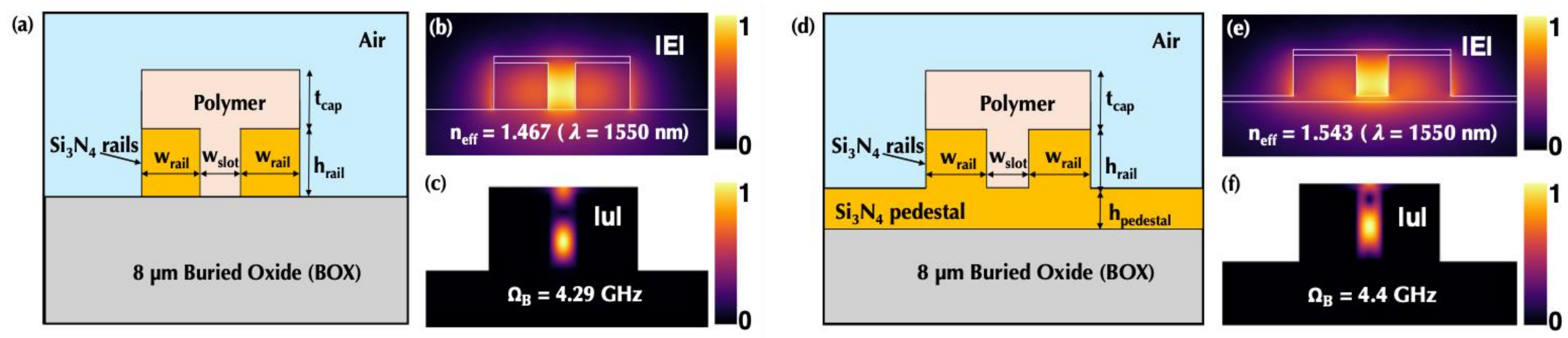
1) Low optical loss at 1550 nm
2) Soft acoustic properties
3) Enhanced photo-elastic response
4) Compatible with spin coating or 2PP fabrication

Material	n_{1550}	ρ_m (kg m^{-3})	Y (GPa)	p_{12}	FM
BCB [2]	1.535	970	2.5-3.0	0.30	Spin Coat
PMMA [3]	1.48	1300	2-5	0.297	,,
CYTOP [4]	1.334	2030	1.5	0.108	,,
IP-Dip [5]	1.527	1200	2.91	0.01	2PP

ρ_m – Mass density, Y– Young modulus, p_{12} – Photo-elastic coefficient, FM: Fabrication method, 2PP: Two photon-polymerization

5. Waveguide Concept & Numerical Modeling

C1: Conventional Slot WG — **C2: Pedestal-Supported Slot WG**



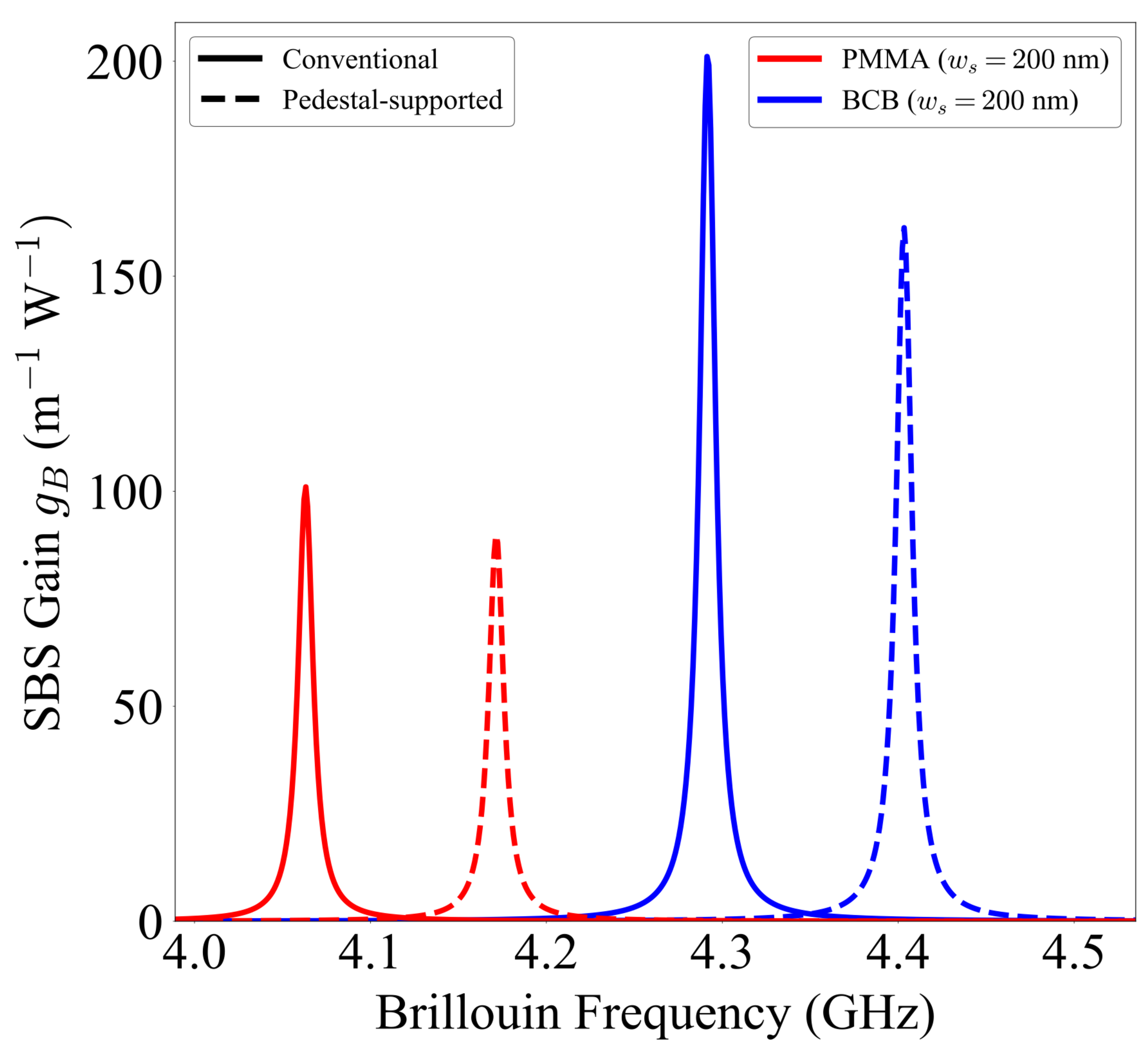
Optimized WG Parameter	Unit (nm)
rail width (w_{rail})	800
rail height (h_{rail})	400
cap thickness (t_{cap})	85
pedestal height ($h_{pedestal}$)	85
slot width (w_{slot})	(100-200)

- Two slot waveguide configurations: C1 & C2
- w_{slot} was optimized individually for each polymer
- Optical and acoustic eigenmodes were analyzed
- Backward SBS simulations in NumBAT

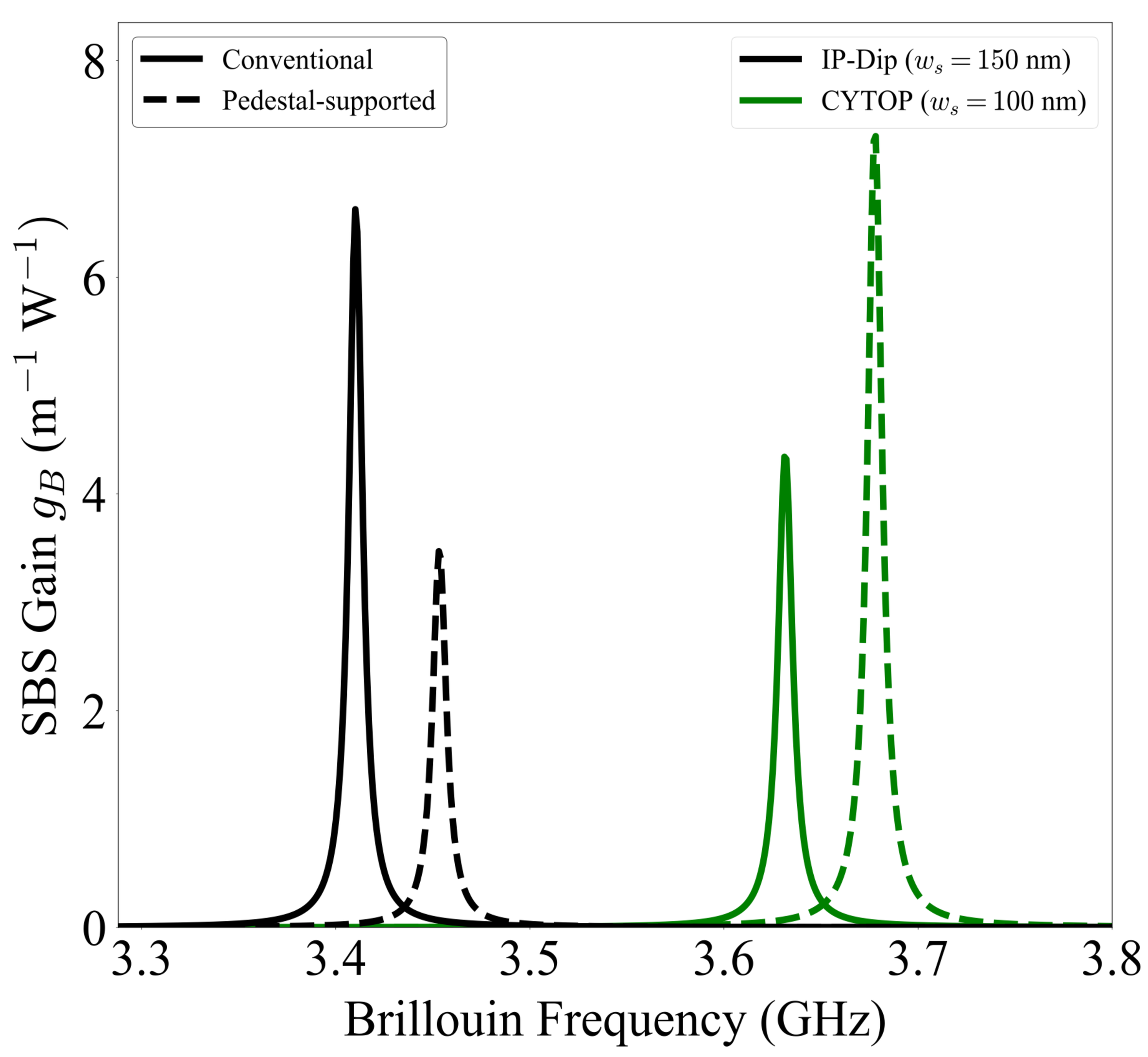
➢ **Mode profiles shown for optimized BCB design**

6. Polymer-Enhanced SBS Performance

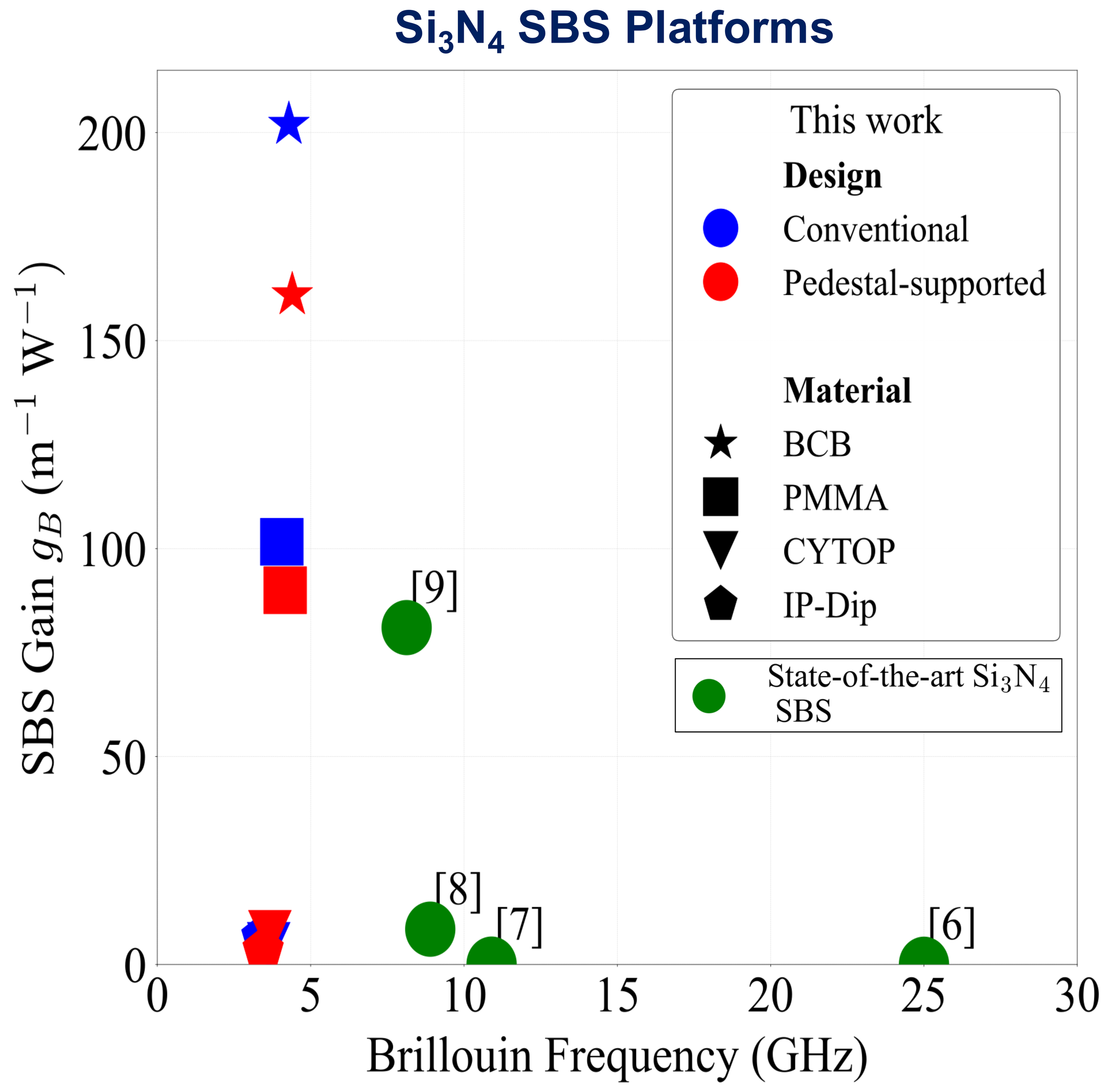
(a) Higher-gain Polymers



(b) Lower-gain Polymers



(c) Comparison with State-of-the-Art Si_3N_4 SBS Platforms



Peak SBS gain: ~ 202 W^{-1} m^{-1}

≈ 2.5× higher than the reported Si_3N_4 SBS waveguide

Key Takeaways

✓ Slot-material selection plays a critical role in SBS enhancement.

✓ BCB yields the highest SBS gain.

✓ PMMA provides moderate SBS gain (~101 W^{-1} m^{-1}).

✓ CYTOP and IP-Dip exhibit significantly poor SBS performance.

✓ Pedestal support modifies acoustic confinement and Brillouin frequency.

❑ **Polymer engineering offers an effective approach to enhance SBS in Si_3N_4 slot waveguides**

7. Outlook

✓ Fabrication of the proposed waveguide designs.

✓ Experimental validation of the numerical gain predictions.

✓ Towards compact on-chip Brillouin photonic devices.

8. References

Scan the QR code for full reference & additional details

9. ACKNOWLEDGMENT

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